P.4 Inverse Functions

Name _____ Period ____ Date _____

AP Calculus P.4 (30-minute plan) · Student sheet

I can…

- decide whether a function is one-to-one.
- find and verify the inverse of a function.
- graph an inverse function and state its domain and range.

Vocabulary · fill in the missing word

One-to-one function — a function in which any two different inputs _____ to two different outputs

Horizontal-line Test — f is _____ if every horizontal line intersects its graph at most once

Inverse function — f^{-1}, where $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$; the _____ of f^{-1} is the range of f

Part 1 · Notes and examples (fill in each note, then work the example)

Horizontal-line Test: if every horizontal line intersects the graph of f at most _____, then f is one-to-one.

Example 1

Is each function one-to-one?

a. $\{(-3, 5), (-2, 9), (-1, 2), (0, 11), (1, -5)\}$

b. $\{(-2, 1), (-3, 2), (-10, 0), (1, 9), (2, 1)\}$

Graphs: the graph of f^{-1} is the reflection of the graph of f about the line _____.

Inverse: $f^{-1}(f(x)) = x$ for x in the domain of f, and $f(f^{-1}(x)) = x$ for x in the domain of _____.

Example 2

Show that $f(x) = 3x + 4$ and $g(x) = (\frac{1}{3})(x - 4)$ are inverses.

Finding f^{-1}: write $y = f(x)$, _____ x and y, and solve for y.

Example 3

Find the inverse of $f(x) = \frac{2x + 3}{x + 2}$ and give the domain and range of f and f^{-1}.

Part 2 · Partner practice

Partner practice

Find the inverse of $f(x) = \sqrt[3]{(x + 10)}$.